

Task 2.2 Project Planning and Sourcing Data with an API

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Part 1: Dashboard Plan for Citi Bike Analysis

Dashboard Structure

The dashboard will have the following sections:

1. Introduction Page

- Brief overview of the project objectives.
- Explanation of data sources (CitiBike Trip Data + NOAA Weather Data).

2. Top 10 Most Popular Start Stations

- Bar Chart visualization.

3. Monthly Trips vs Temperature Trends

- Dual Line Chart (Number of Trips and Average Temperature over months).

4. Most Popular Bike Trips (Station-to-Station Routes)

- Interactive Kepler.gl map showing most common routes.

5. Station Distribution Across NYC

- Interactive Kepler.gl map showing all bike stations.

6. Recommendations and Strategic Insights

- Actionable advice based on the findings.

GitHub Repository

Repository Name: CitiBike_NYC_2022_Analysis

Repo Link: https://github.com/SamarjitSingh-hub/CitiBike_NYC_2022_Analysis

Key Research Questions and How They Will Be Visualized

Research Question	Visualization Approach	Purpose
What are the most popular start stations in New York City?	Bar Chart (Top 10 Stations by trip count)	Identify high-demand locations for possible expansion.
Which months had the highest number of trips?	Line Chart (Monthly trip volume)	Detect seasonal demand patterns.
Is there a correlation between weather and trip volume?	Line Chart (Overlay Monthly Avg Temp vs Trip Count)	Understand how weather impacts usage.
Which station-to-station routes are most popular?	Kepler.gl Map (Trip start and end points)	Spot common travel patterns and bottlenecks.
Are the bike stations evenly distributed across NYC?	Kepler.gl Map (Bike station locations)	Identify underserved regions needing new stations.